NIKUL PATEL

pnikul165@gmail.com | LinkedIn [n1kul] | +1(315) 603-7990 | Open to Relocate

EDUCATION

Clarkson University *Potsdam, NY*
Double Major: Aerospace and Mechanical Engineering | Minor: Robotics | Honors Program Expected Graduation: 2028
Clubs and Organizations: CUSA | SASE | Underwater Robotics GPA: 3.986

PROFESSIONAL EXPERIENCE

Undergraduate Researcher Dec 2025 - Present
PI: Prof. Gordon Li *Potsdam, NY*
- Developing a physics-based architecture for autonomous in-space assembly, integrating robotic agents and tether-supported modular structures for autonomous module replacement
- Modeling multi-agent dynamics and autonomous control strategies in MATLAB/Simulink to simulate assembly operations and evaluate system behavior

CubeSat Digital Twin Jan 2026 - Present
Independent Engineering Project (Scan QR above for other projects) *Potsdam, NY*
- Developing an end-to-end spacecraft digital twin by integrating SOLIDWORKS CAD-derived mass properties and inertia tensors with a MATLAB physics engine modeling orbital mechanics, environmental perturbations, and quaternion-based rigid-body dynamics
- Implementing closed-loop ADCS with EKF, PD attitude control, and reaction wheel/magnetorquer actuator management; demonstrated autonomous nadir-pointing slew and disturbance rejection in a simulation run

Design Build Captain Aug 2025 - Present
Underwater Robotics Team *Potsdam, NY*
- Designing and 3D-printing custom mechanical fixtures and supports in SolidWorks and Onshape to improve robotic subsystem integration, structural reliability, and assembly performance
- Collaborating with electrical and controls teams to integrate robotic subsystems, verify dimensional interfaces, and implement DFMA improvements through iterative design and testing

Co-Op Student Jan 2026 - July 2026
Baldwin Richardson Foods *Rochester, NY*
- Supported Lean Six Sigma initiatives through 2 Kaizen events and 10+ root cause analyses, implementing corrective actions to improve operational efficiency and process reliability
- Led a DMAIC-based LOTO standardization project, identifying process gaps and revising 50+ work instructions and compliance documents to improve operational consistency and safety

LEADERSHIP AND ACADEMIC ENGAGEMENT

*Most Recently held Position

Teaching Assistant | Honors MATLAB (HP102/103) | Clarkson University *Aug 2025 - Present*
- Co-managing a fully TA-led honors course, coordinating lectures, curriculum, assignments, grading, and office hours

Engagement Chair and Senator | Student Government (CUSA) | Clarkson University *Sept 2025 - Present*
- Leading campuswide engagement initiatives and advocating student interests through Student Government governance

President* | SASE (Society of Asian Scientists and Engineers) | Clarkson Chapter *Sept 2024 - Present*
- Leading chapter operations and a 12-member executive board to advance organizational goals and member engagement

Student Manager* | Undergraduate Admissions | Clarkson University *Sept 2024 - Present*
- Supervising a 30-member team of tour guides overseeing admissions outreach and prospective-student engagement

SKILLS

*Certified Skill

Design & Prototyping: SolidWorks*, Onshape, AutoCAD, GD&T*, DFMA*, Technical Drawings & Documentation, Rapid Prototyping, FDM 3D Printing*, Custom Fixture Design
Analysis & Simulation: ANSYS*, FEA, Thermal Analysis, Hardware-in-the-Loop (HIL) Simulation
Robotics & Hardware: Mechatronics, Electromechanical Systems, ROS, Arduino Embedded Systems, Pneumatic Systems, Sensor Integration, Actuators & Motors
Programming & Software: Python*, MATLAB/Simulink*, Deep Learning & CNNs (PyTorch/TensorFlow)
Continuous Improvement & Operations: Lean Six Sigma*, DMAIC, Kaizen, Root Cause Analysis, KPI Analysis

AWARDS

Presidential Scholar: Academic Excellence | *Clarkson University* *All Semesters*
SASE Rising Leader: Recognition for strong leadership skills and proactive initiative | *Clarkson University* *April 2025*